# Psychometric report: Statistical Reasoning of 15-10-2025

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## Summary

According to the *Guideline Assessment Analysis*, 25 out of 37 items (67.57%) show one or more psychometric discrepancies. These questions may provide useful insights when reviewed.

- 1 item(s) are very difficult
- 17 item(s) are very easy
- 8 item(s) show low discrimination
- 0 item(s) show negative discrimination
- 1 item(s) show negative item-rest correlation
- 8 item(s) reduce internal consistency

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## 1. Descriptives

This section provides an overview of the assessment results and key characteristics of the score distribution.

### 1.1 Statistics about achieved scores

The assessment included 128 participants. The table below summarizes key descriptive statistics of their achieved scores.

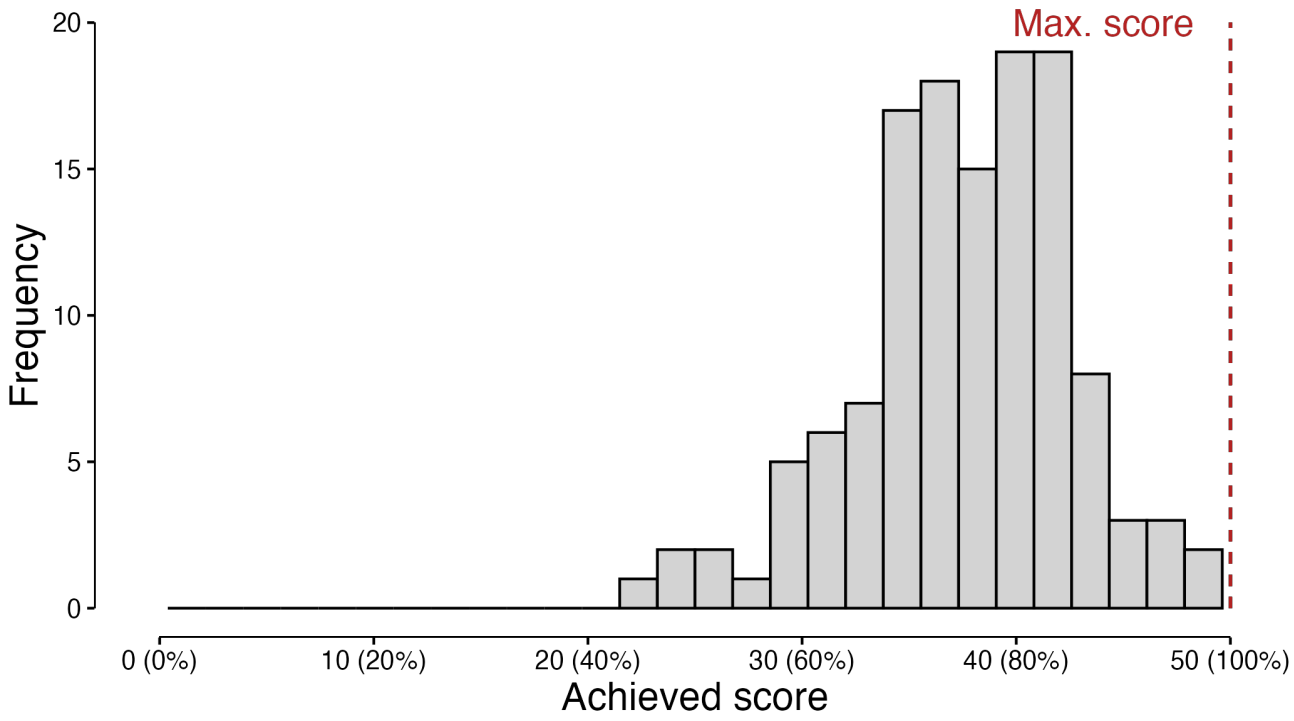
- The average score of 37.566 (75.132%) represents the central tendency of the scores and reflects the typical performance level.
- The median of 38.5 (77%) is the middle score when all participants are ordered from lowest to highest.
- The standard deviation of 5.038 (10.077%) measures the spread of scores around the average. Higher values indicate greater variability, while lower values indicate scores clustered near the mean.
- The skewness of -0.506 indicates that the distribution is negatively skewed, with a tendency toward higher scores. Positive skew reflects more low scores and fewer high scores, while negative skew indicates the opposite.

- The kurtosis of 0.239 means the distribution is approximately normal in peakedness, indicating a balanced spread of scores around the mean. Higher values indicate a stronger concentration of scores near the mean with more extreme values, while lower values indicate a flatter distribution.

| Statistic | Value |
|---|---|
| Number of participants | 128 |
| Maximum possible score | 50 (100%) |
| Minimum achieved score | 22.75 (45.5%) |
| Maximum achieved score | 49 (98%) |
| Average achieved score | 37.566 (75.132%) |
| Median achieved score | 38.5 (77%) |
| Standard deviation | 5.038 (10.077%) |
| Skewness | -0.506 |
| Kurtosis | 0.239 |

## 1.2 Distribution of achieved scores

The histogram below visualizes the distribution of participant scores. The horizontal axis represents score ranges, and the vertical axis indicates the number of participants within each range.



- Most participants scored in the upper range (>70%), indicating that this range was most effective in differentiating participant performance.
- Look out for peaks near the maximum or minimum possible scores, as this may indicate ceiling effects (many very high scores) or floor effects (many very low scores), which can impact the assessment's ability to distinguish between stronger and weaker participants.
- Multiple peaks in the distribution may indicate heterogeneous participant groups with different performance levels.

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# 2. Assessment analysis

An assessment analysis uses statistical indicators to evaluate the quality of the assessment (items). These metrics should be interpreted in the context of the assessment purpose, participant population, and sample size.

## 2.1 Assessment statistics

The table below presents key evaluation metrics for the overall assessment. Cell colors follow the thresholds defined in the *Guideline Assessment Analysis*.

- *Average P (Difficulty)*: A value near 0 indicates a very difficult assessment, whereas a value near 1 indicates a very easy one. An assessment with moderate difficulty (typically between 0.2 and 0.8) tends to provide the greatest discriminatory power.
- *Average RIT and RIR (Discrimination)*: These statistics measure the extent to which items differentiate between higher- and lower-performing participants. Values below 0.2 (red) indicate weak discrimination, values between 0.2 and 0.3 (orange) indicate moderate discrimination, and values above 0.3 (green) indicate strong discrimination.
- *Cronbach's alpha (Internal Consistency)*: This is the internal consistency of the assessment. A value above 0.7 (green) generally indicates acceptable reliability. Lower values may suggest inconsistent items or items that contribute little to the measurement of the intended construct. A very high value (>0.9) may indicate redundancy among items.

| Average P | Average RIT | Average RIR | Cronbach's alpha |
|---|---|---|---|
| 0.737 | 0.309 | 0.233 | 0.728 |

## 2.2 Item statistics

This table summarizes key evaluation metrics for each individual item. Cell colors follow the thresholds defined in the *Guideline Assessment Analysis*.

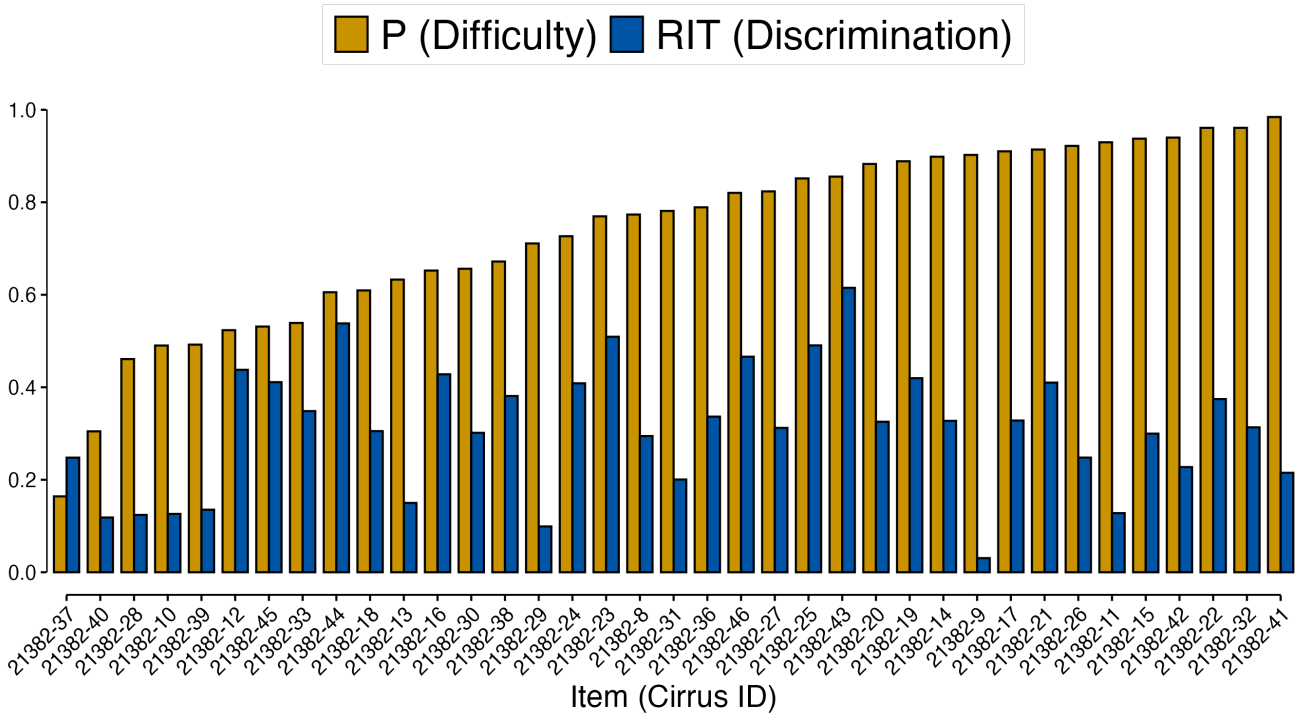
- *P (Item Difficulty)*: The P-value represents the proportion of participants who answered the item correctly. Values near 0 indicate very difficult items, while values near 1 indicate very easy items.
- *RIT (Item-Total Correlation)*: RIT represents the correlation between an item score and the total score. Higher values indicate that the item aligns well with overall performance. Negative values require immediate review, as they indicate that lower-performing participants answered the item correctly more often than higher-performing participants.
- *RIR (Item-Rest Correlation)*: RIR represents the correlation between an item and the total score excluding that item. Interpretation follows the same thresholds as RIT and provides a similar indication of item discrimination.
- *Alpha-if-deleted*: This statistic indicates the change in Cronbach's alpha if the item were removed from the assessment. An increase in alpha (red) suggests that the item may reduce internal consistency.

| Item (Cirrus ID) | Mean | SD | P | RIT | RIR | Alpha-if-deleted |
|---|---|---|---|---|---|---|
| 21382-8 | 2.320 | 0.790 | 0.773 | 0.295 | 0.143 | 0.734 |
| 21382-9 | 1.805 | 0.282 | 0.902 | 0.031 | -0.025 | 0.732 |
| 21382-10 | 0.490 | 0.415 | 0.49 | 0.126 | 0.044 | 0.731 |
| 21382-11 | 0.930 | 0.257 | 0.93 | 0.128 | 0.077 | 0.728 |
| 21382-12 | 0.523 | 0.501 | 0.523 | 0.438 | 0.352 | 0.714 |
| 21382-13 | 0.633 | 0.484 | 0.633 | 0.15 | 0.054 | 0.732 |
| 21382-14 | 0.898 | 0.303 | 0.898 | 0.327 | 0.272 | 0.721 |
| 21382-15 | 0.938 | 0.243 | 0.938 | 0.3 | 0.255 | 0.723 |
| 21382-16 | 1.305 | 0.647 | 0.652 | 0.428 | 0.315 | 0.716 |
| 21382-17 | 0.910 | 0.165 | 0.91 | 0.328 | 0.298 | 0.723 |
| 21382-18 | 0.609 | 0.490 | 0.609 | 0.305 | 0.213 | 0.723 |
| 21382-19 | 1.777 | 0.429 | 0.889 | 0.42 | 0.346 | 0.716 |
| 21382-20 | 0.883 | 0.323 | 0.883 | 0.325 | 0.266 | 0.721 |
| 21382-21 | 1.828 | 0.254 | 0.914 | 0.41 | 0.367 | 0.719 |
| 21382-22 | 1.922 | 0.238 | 0.961 | 0.374 | 0.333 | 0.721 |
| 21382-23 | 1.539 | 0.675 | 0.77 | 0.509 | 0.4 | 0.709 |
| 21382-24 | 0.727 | 0.447 | 0.727 | 0.409 | 0.331 | 0.716 |
| 21382-25 | 1.703 | 0.573 | 0.852 | 0.491 | 0.397 | 0.71 |

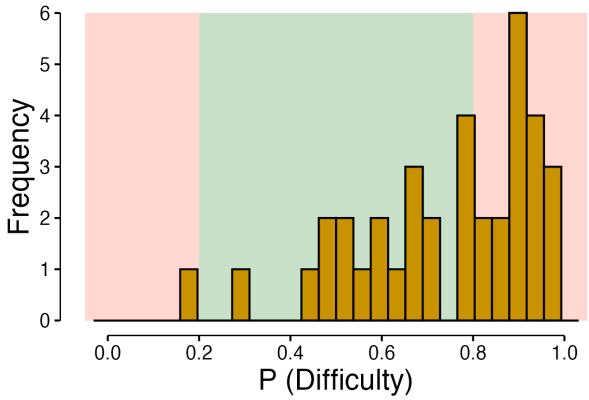
| | | | | | | |
|---|---|---|---|---|---|---|
| 21382-26 | 0.922 | 0.269 | 0.922 | 0.248 | 0.197 | 0.724 |
| 21382-27 | 0.824 | 0.268 | 0.824 | 0.312 | 0.263 | 0.722 |
| 21382-28 | 0.461 | 0.500 | 0.461 | 0.124 | 0.025 | 0.734 |
| 21382-29 | 0.711 | 0.455 | 0.711 | 0.099 | 0.009 | 0.734 |
| 21382-30 | 0.656 | 0.477 | 0.656 | 0.301 | 0.212 | 0.723 |
| 21382-31 | 0.781 | 0.415 | 0.781 | 0.201 | 0.12 | 0.728 |
| 21382-32 | 0.961 | 0.195 | 0.961 | 0.313 | 0.278 | 0.723 |
| 21382-33 | 0.539 | 0.500 | 0.539 | 0.348 | 0.257 | 0.72 |
| 21382-36 | 0.789 | 0.410 | 0.789 | 0.337 | 0.262 | 0.72 |
| 21382-37 | 0.164 | 0.372 | 0.164 | 0.248 | 0.177 | 0.725 |
| 21382-38 | 0.672 | 0.471 | 0.672 | 0.381 | 0.297 | 0.718 |
| 21382-39 | 0.492 | 0.502 | 0.492 | 0.135 | 0.036 | 0.734 |
| 21382-40 | 0.305 | 0.462 | 0.305 | 0.118 | 0.027 | 0.733 |
| 21382-41 | 0.984 | 0.125 | 0.984 | 0.215 | 0.191 | 0.726 |
| 21382-42 | 0.940 | 0.199 | 0.94 | 0.227 | 0.189 | 0.725 |
| 21382-43 | 1.711 | 0.424 | 0.855 | 0.615 | 0.558 | 0.705 |
| 21382-44 | 1.211 | 0.695 | 0.605 | 0.538 | 0.429 | 0.706 |
| 21382-45 | 1.062 | 0.707 | 0.531 | 0.411 | 0.285 | 0.719 |
| 21382-46 | 1.641 | 0.498 | 0.82 | 0.466 | 0.383 | 0.712 |

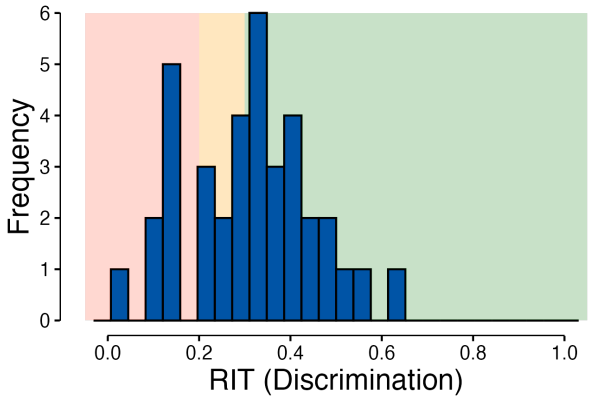
The plot below displays item difficulty (P-value) in relation to item discrimination (RIT).

- Items that are very easy or very difficult and show low discrimination may provide limited measurement information and may warrant revision.
- Items with moderate difficulty and strong discrimination generally contribute most effectively to reliable measurement.




The item difficulty (P-value) and item discrimination (RIT) are shown again in the figures below, with colored areas representing thresholds.

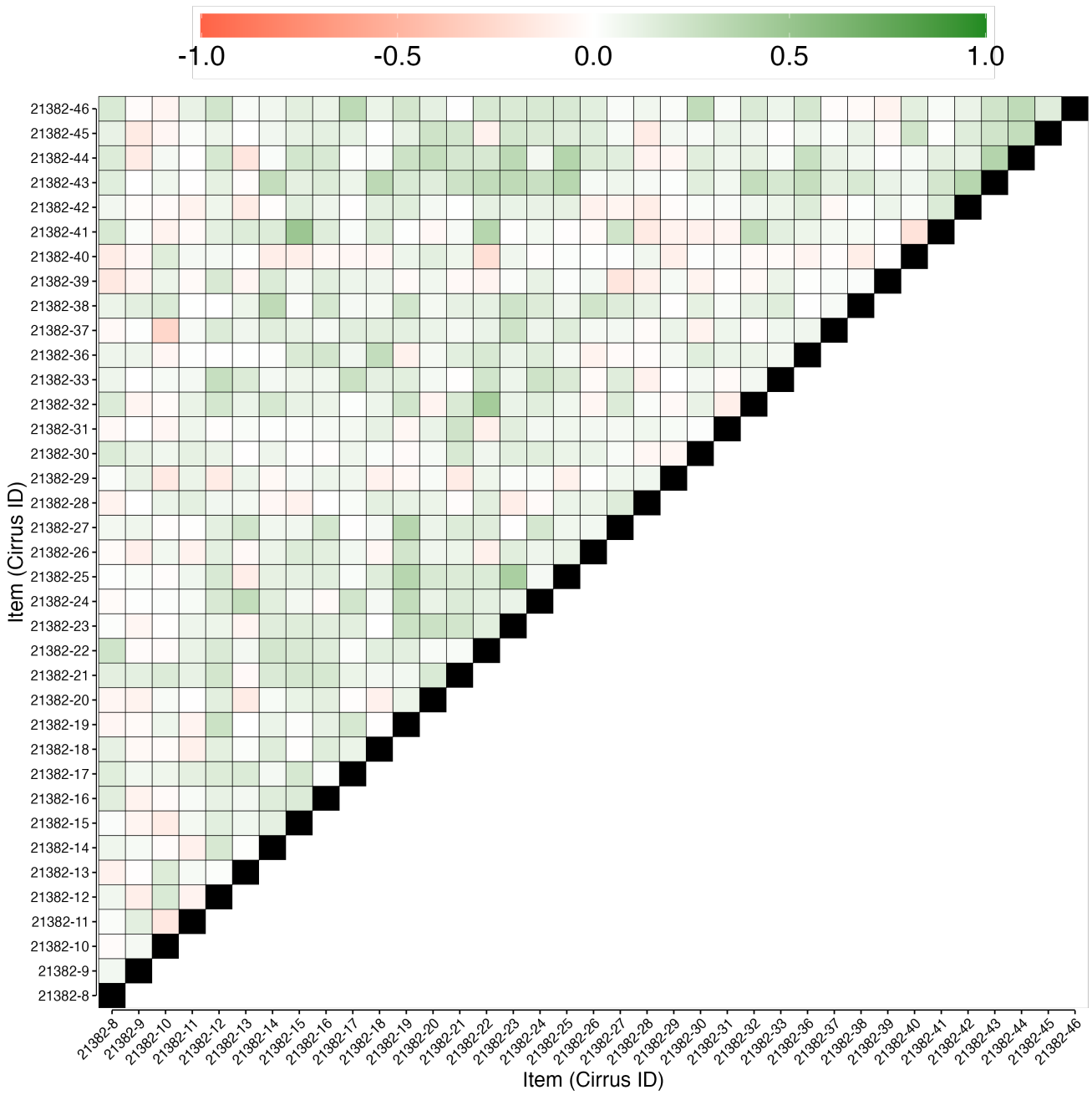



## 2.3 Item correlations

This heatmap below shows correlations between all assessment items.

- Items showing strong positive correlations (>0.6) may measure very similar content and should be reviewed for potential redundancy.
- Items with negative correlations with multiple other items should be reviewed for potential scoring errors or content misalignment.




The table below below shows the five strongest positive and five strongest negative correlations. Red values indicate extremely strong correlations (>0.6) and negative correlations.

| Item (Cirrus ID) | Item (Cirrus ID) | Correlation |
|---|---|---|
| 21382-15 | 21382-41 | 0.488 |
| 21382-22 | 21382-32 | 0.443 |
| 21382-23 | 21382-25 | 0.417 |
| 21382-43 | 21382-44 | 0.369 |
| 21382-25 | 21382-44 | 0.366 |
| 21382-10 | 21382-37 | -0.27 |
| 21382-22 | 21382-40 | -0.211 |
| 21382-40 | 21382-41 | -0.19 |
| 21382-27 | 21382-39 | -0.167 |
| 21382-13 | 21382-44 | -0.166 |

# 3. Items flagged for review

The table below summarizes the 25 items that may require review due to psychometric concerns. Each flagged item is listed with the specific issue(s) detected and an explanation for why it may warrant attention. The table is ordered so that the most relevant items (i.e., with the most issues) are shown first.

| Item (Cirrus ID) | Issue | Explanation |
|---|---|---|
| 21382-9 | Very easy item; Low discrimination; Reduces test reliability; Negative item-rest correlation | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. Item-total correlation is below 0.20, which may incidate a weak ability to distinguish between stronger and weaker participants. Item removal increases Cronbach's alpha, which may indicate that the item does not align well with the construct measured by the assessment. Item correlates negatively with the rest of the assessment, which may indicate a potential scoring error or content misalignment. |
| 21382-10 | Low discrimination; Reduces test reliability | Item-total correlation is below 0.20, which may incidate a weak ability to distinguish between stronger and weaker participants. Item removal increases Cronbach's alpha, which may indicate that the item does not align well with the construct measured by the assessment. |
| | | More than 80% of participants answered correctly, |

| | | |
|---|---|---|
| 21382-11 | Very easy item; Low discrimination | meaning the item provides limited discrimination between participants. Item-total correlation is below 0.20, which may incidate a weak ability to distinguish between stronger and weaker participants. |
| 21382-13 | Low discrimination; Reduces test reliability | Item-total correlation is below 0.20, which may incidate a weak ability to distinguish between stronger and weaker participants. Item removal increases Cronbach's alpha, which may indicate that the item does not align well with the construct measured by the assessment. |
| 21382-28 | Low discrimination; Reduces test reliability | Item-total correlation is below 0.20, which may incidate a weak ability to distinguish between stronger and weaker participants. Item removal increases Cronbach's alpha, which may indicate that the item does not align well with the construct measured by the assessment. |
| 21382-29 | Low discrimination; Reduces test reliability | Item-total correlation is below 0.20, which may incidate a weak ability to distinguish between stronger and weaker participants. Item removal increases Cronbach's alpha, which may indicate that the item does not align well with the construct measured by the assessment. |
| 21382-39 | Low discrimination; Reduces test reliability | Item-total correlation is below 0.20, which may incidate a weak ability to distinguish between stronger and weaker participants. Item removal increases Cronbach's alpha, which may indicate that the item does not align well with the construct measured by the assessment. |
| 21382-40 | Low discrimination; Reduces test reliability | Item-total correlation is below 0.20, which may incidate a weak ability to distinguish between stronger and weaker participants. Item removal increases Cronbach's alpha, which may indicate that the item does not align well with the construct measured by the assessment. |
| 21382-8 | Reduces test reliability | Item removal increases Cronbach's alpha, which may indicate that the item does not align well with the construct measured by the assessment. |
| 21382-14 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-15 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |

| | | |
|---|---|---|
| 21382-17 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-19 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-20 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-21 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-22 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-25 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-26 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-27 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-32 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-37 | Very difficult item | Less than 20% of participants answered correctly, suggesting the item may be ambiguous, miskeyed, or cover material not sufficiently taught. |
| 21382-41 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-42 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| 21382-43 | Very easy item | More than 80% of participants answered correctly, meaning the item provides limited discrimination between participants. |
| | Very easy | More than 80% of participants answered correctly, |

| | | |
|---|---|---|
| 21382-46 | item | meaning the item provides limited discrimination between participants. |